

8-bit Left Shift Register



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ED221: Digital IC Design Tape in & Tape out Lab

CMOS Implementation of an 8-Bit Left Shift Register in Cadence

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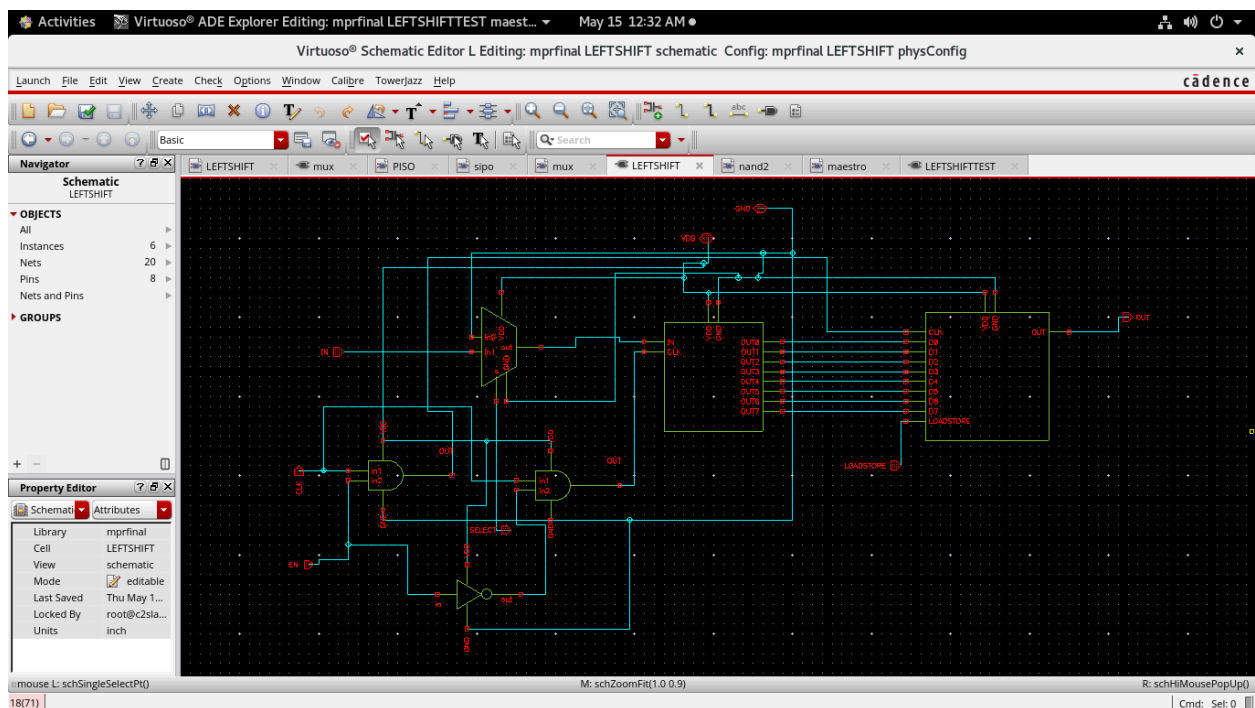
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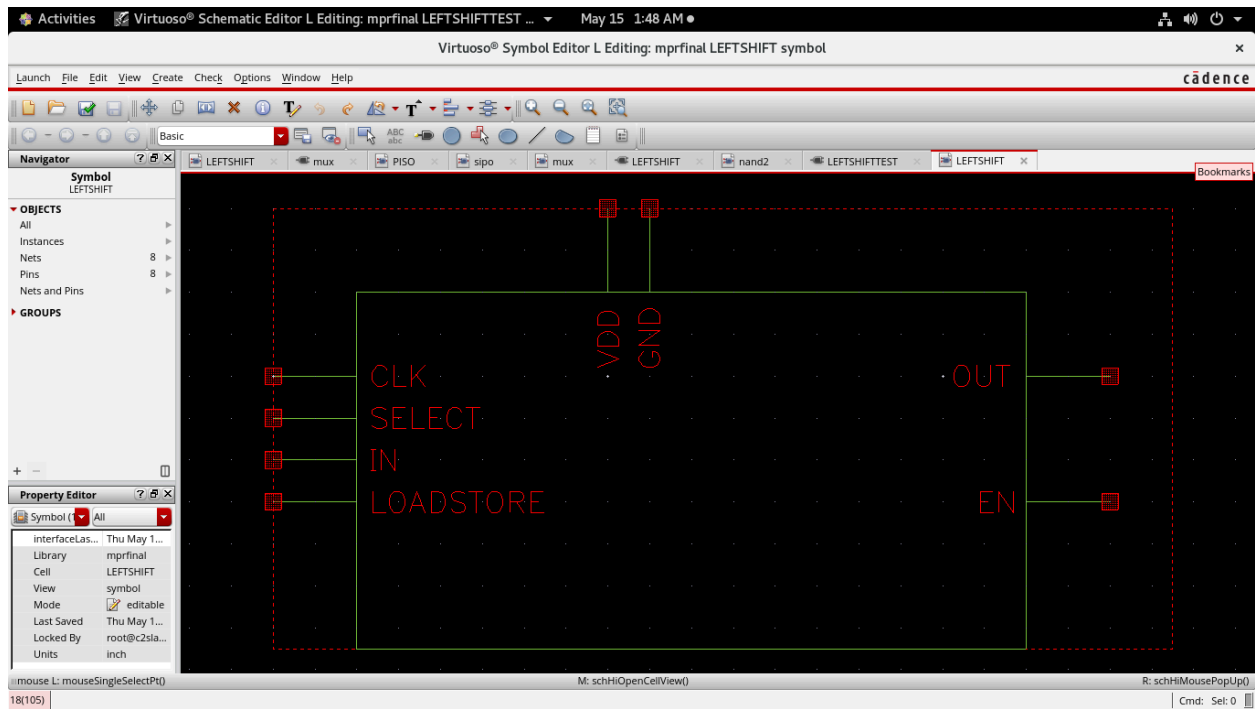
1 | Introduction

The 8-bit Left Shift Register circuit is implemented using a combination of Parallel-In Serial-Out (PISO), Serial-In Serial-Out (SISO) registers, multiplexers, and logic gates to perform logical left-shift operations on 8-bit data. The design operates at a nominal supply voltage of 1.8V and is implemented and simulated in Cadence Virtuoso. The register supports synchronized clock-driven shifting, serial data transfer, and logical left-shift functionality.

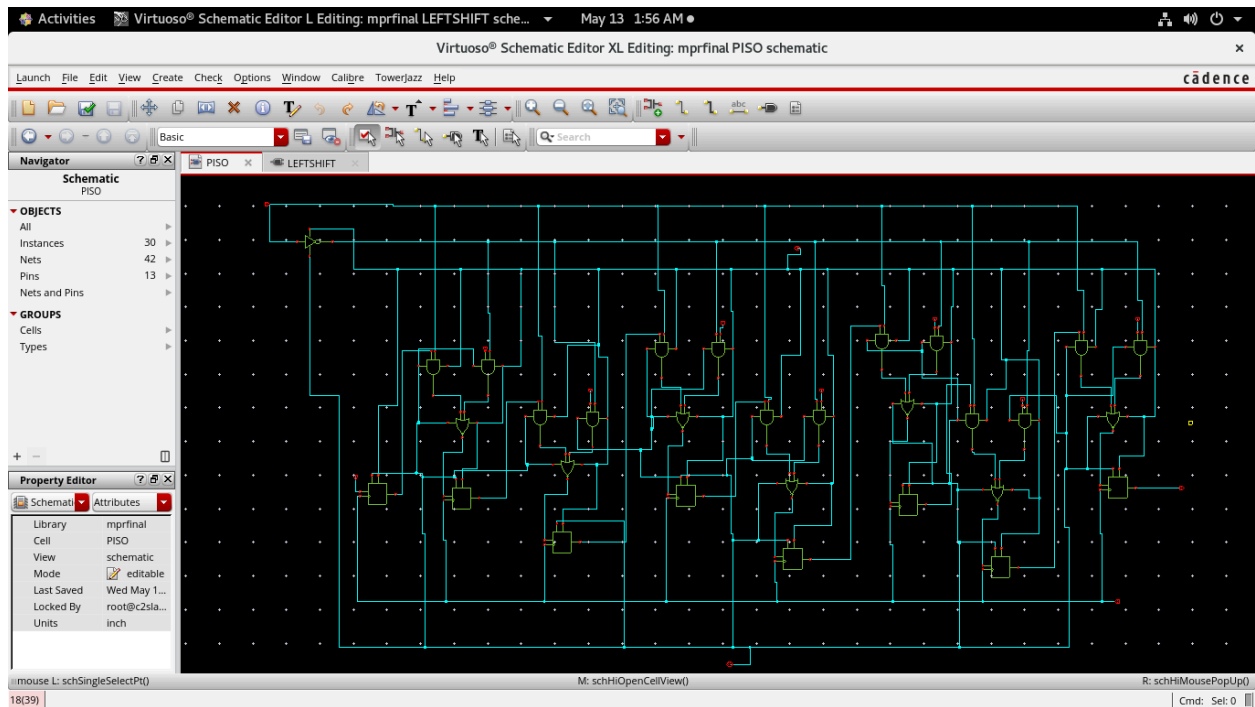
2 | Circuit Schematic



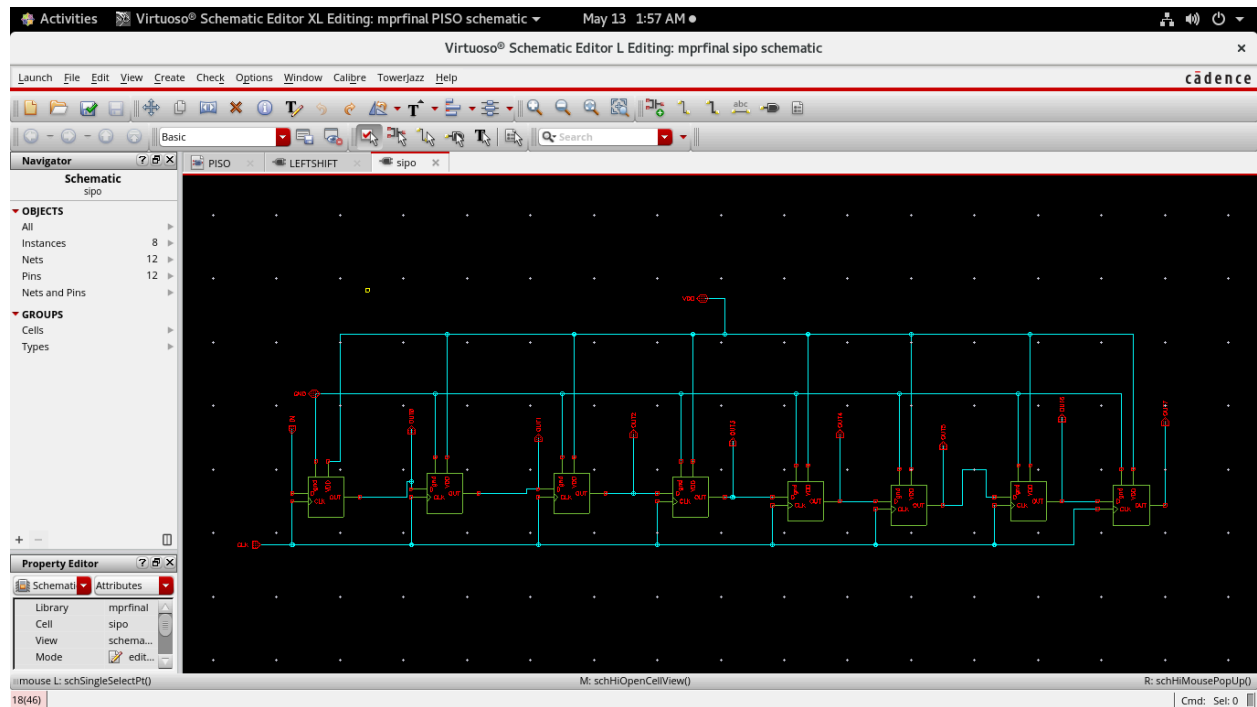
SYMBOL



PISO



SIPO



3 | Pin Configuration

Pin	Description
VDD	Supply voltage (1.8V)
GND	Ground (0V)
CLK	Clock input
IN	Serial data input (MSB first)
LOADSTORE	Control signal for load/shift operation
SELECT	Shift operation select input
EN	Enable signal for clock operation
OUT	Serial Output (MSB first)

4 | Delays of Components used

Components	Delay
AND Gate	227.7 ps
OR Gate	174.05 ps
MUX	189.72 ps
LATCH	428.395 ps
D Flip-Flop	506.35 ps

- 8-BIT LEFT SHIFT REGISTER DELAY(DATA = 11111111): 0.74 ns

5 | Power Characteristics (DATA = 11111111)

Parameter	Symbol	Value	Unit
Average Current	I_{avg}	64.18	μA
Peak Current	I_{peak}	2.92	mA
Average Power	P_{avg}	115.524	μW

6 | Calculations

- Average Power Consumption:

$$\text{Average Power} = V_{DD} \times I_{avg}$$

$$\text{Average Power} = 1.8 \text{ V} \times 64.18 \mu A$$

$$\text{Average Power} = 115.524 \mu W$$

- Energy Consumption:

$$\text{Energy} = T_{sim} \times P_{avg}$$

$$\text{Energy} = 500 \text{ ns} \times 115.524 \mu W$$

$$\text{Energy} = 57.762 \mu J$$

7 | Functional Description

7.1. SIPO

- SIPO is used for serial data input and left-shift data transfer operations. The shifted data is obtained serially with the MSB appearing first at the output.

7.2. PISO

- PISO is used for parallel data loading and serial data output operations. The stored 8-bit data is shifted out serially with the MSB appearing first at the output.
- **LOADSTORE** is the control signal used for load and shift operations. When **LOADSTORE** = 0, the data is loaded into the register, and when **LOADSTORE** = 1, the data is shifted and given serially at output.

7.3. MUX

- MUX is used to select the amount of left-shift operation to be performed on the data. The shift selection is controlled using the **SELECT** input pin.

SELECT	Function
0	Extra zeroes for shifting
1	Data in

7.4. AND

- The AND gates are used to control the clock signals of the SIPO and PISO shift registers using the Enable (EN) signal.
 - For **PISO**, the clock is generated by:

$$CLK_{PISO} = CLK \cdot EN$$

$$CLK_{SIPO} = CLK \cdot ENbar$$

8 | Working

8.1. Stages of Operation of the SIPO–PISO System

- Stage 1: Data Input**
 - The serial input data is applied to the input line.
 - Clock pulses are provided to synchronize the data transfer.
 - At this stage, the system starts receiving bits one by one.
- Stage 2: SIPO Mode ON (Serial-In Parallel-Out)**
 - The SIPO shift register is enabled.
 - With every clock pulse, the incoming serial data shifts through the flip-flops.
 - After all bits are received, the complete data becomes available in parallel form at the SIPO outputs.
 - The SIPO stores and arranges the data properly for the next stage.
- Stage 3: PISO Mode ON (Parallel-In Serial-Out)**
 - Once all parallel data is available from the SIPO, the PISO register is enabled.

-
- The enable signal is ANDed with the clock signal before being applied to the PISO.
 - The PISO loads the parallel data from the SIPO outputs.
 - On each clock pulse, the PISO shifts the data serially and provides serial output bit by bit.

9 | Example stimulation of 7 - Bit Left Shift of 11111111

1. The input data **11111111** is applied through the **IN** pin, and the **SELECT** line of the MUX is kept at logic **1** so that the data can be serially entered into the SIPO register, which receives all 8 bits sequentially.
2. After loading the complete data, the **SELECT** line is changed to logic **0** to perform the 7-bit left-shift operation by inserting seven zeros, due to which the output of the SIPO register becomes:

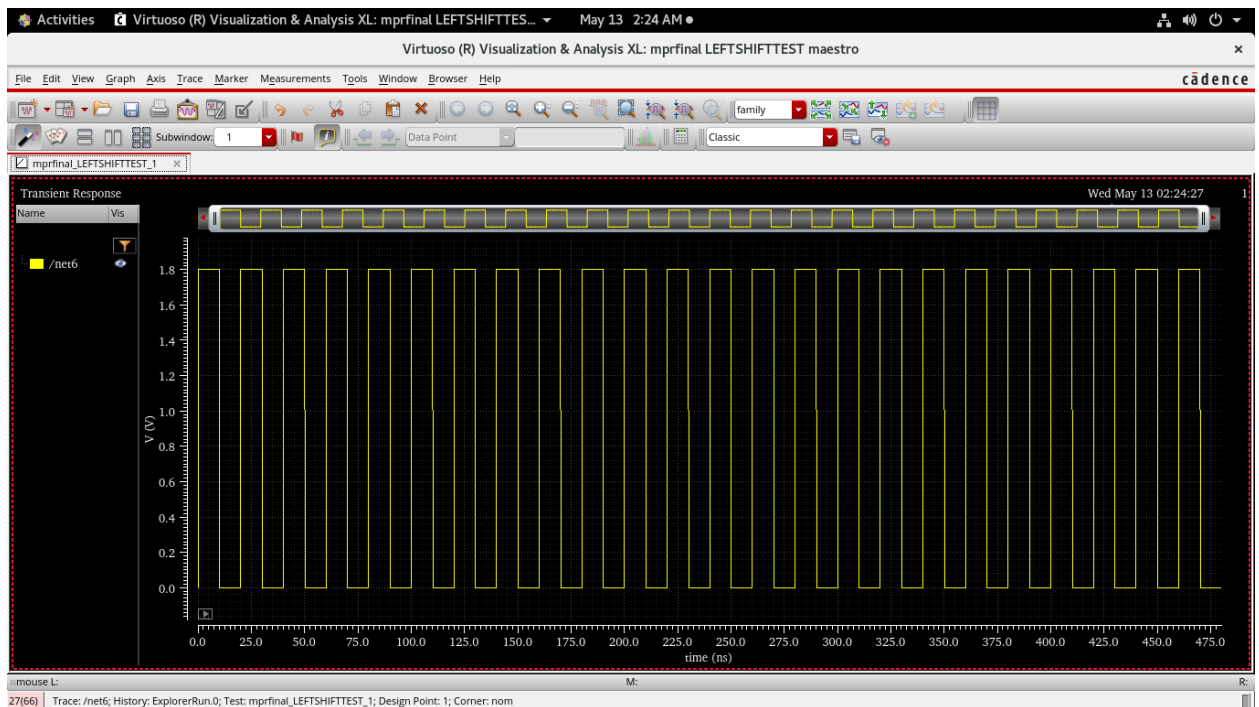
00000001

3. During the shifting operation, the **LOADSTORE** signal remains at logic **1**, so the shifted data is not loaded into the PISO register.
4. After obtaining the shifted output from the SIPO register, the **LOADSTORE** signal is changed to logic **0** to load the data into the PISO register.
5. The PISO register then sends the data serially from **MSB** to **LSB** through the output pin.

Setup:

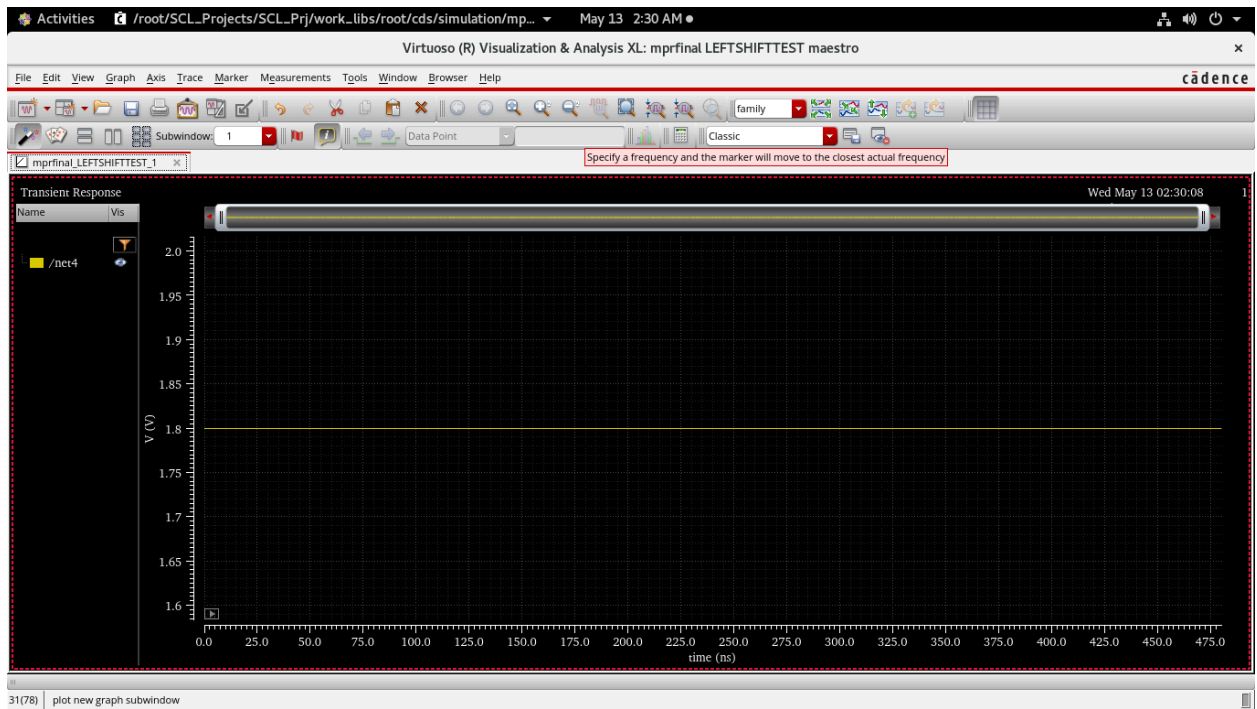
1. Clock:

- The clock signal is configured with a time period of **20 ns** and a duty cycle of **50%**, where the ON time and OFF time are each equal to **10 ns**.



2. Data:

- The data is sampled at the positive edge of the clock. Therefore, the data duration for one bit is equal to one complete clock cycle, i.e., **20 ns**.
- For example, if the input data is **10101010**, the ON time would be **20 ns** and the total cycle time would be **40 ns**. In this design, the input data used is **11111111**; therefore, the input behaves as a constant DC voltage of **1.8 V**, continuously representing logic **1**.



3. Select:

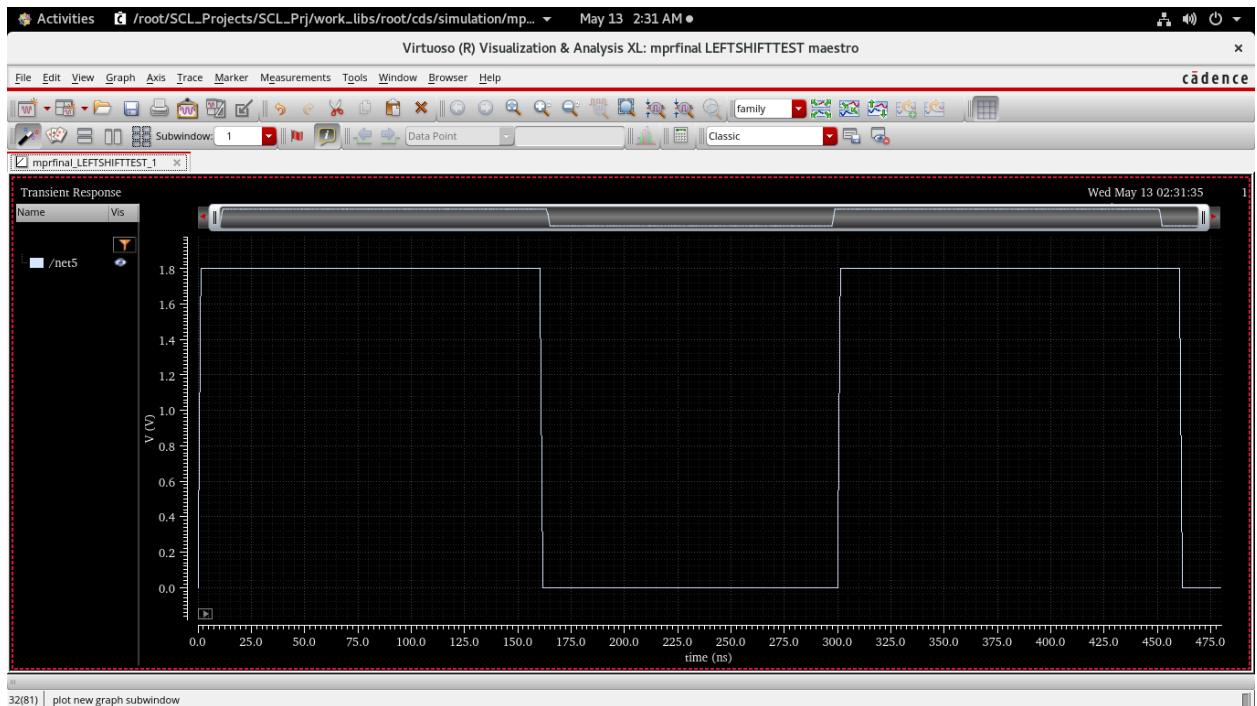
- Initially, the **SELECT** signal is set to logic **1** to allow serial loading of all 8 bits into the SIPO register. Since each bit requires **20 ns**, the total loading time is:

$$8 \times 20 = 160 \text{ ns}$$

- For proper setup timing, the **SELECT** signal is kept high for **159 ns**.

- After loading the data, the **SELECT** signal is changed to logic **0** to perform the left-shift operation. Since the data is shifted by **7 bits**, seven zeros are inserted. Therefore, the required shifting time is:

$$7 \times 20 = 140 \text{ ns}$$



4. ENABLE:

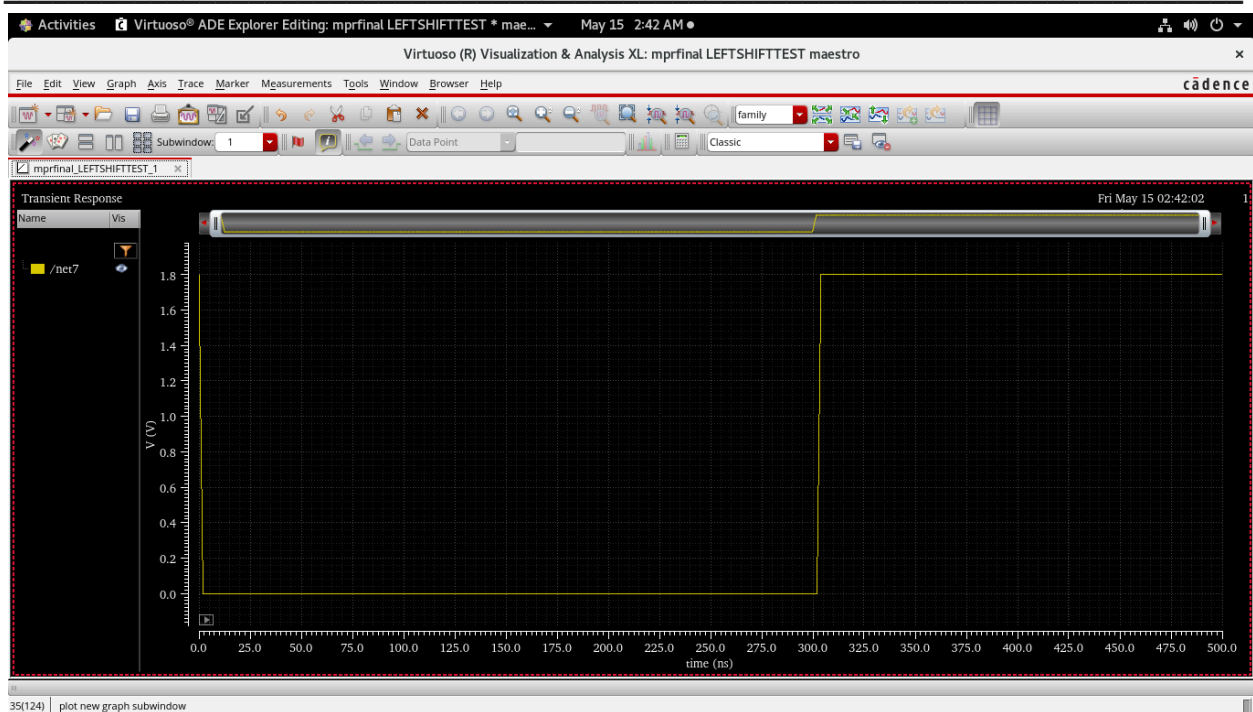
- During the initial loading and shifting operations, the PISO register should not load the data. Therefore, the **ENABLE** signal remains at logic 0 during this duration:

$$159 + 140 \approx 300 \text{ ns}$$

- Then for 8 clock cycles PISO would be ON and will give serial output. Therefore, the **ENABLE** signal remains at logic 1 during this duration:

$$8 * 20 = 160 \text{ ns}$$

- After the shifted output becomes available, the **LOADSTORE** signal is changed to logic 0 for one clock cycle (20 ns) so that the PISO register can load the shifted data.
- Once the data is loaded, the PISO register returns to shift mode, and the **LOADSTORE** signal becomes logic 1 again.



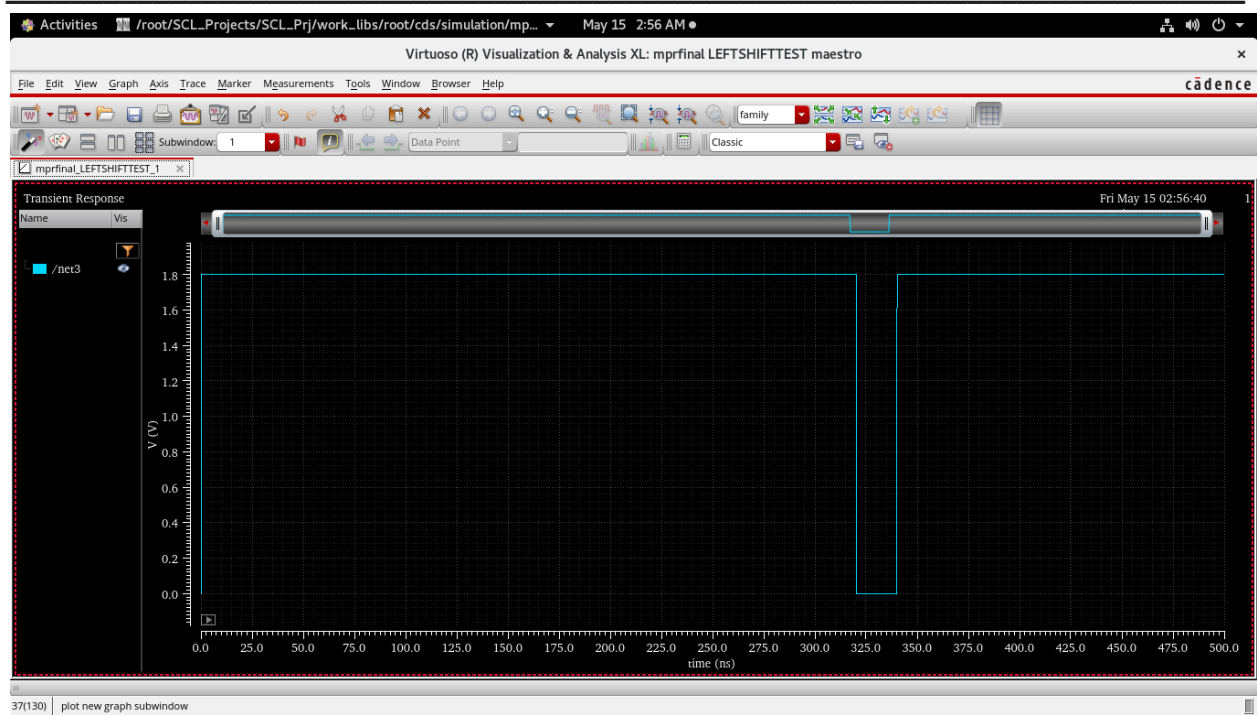
5. LOADSTORE:

- During the initial loading and shifting operations, the PISO register should not load the data. Therefore, the **LOADSTORE** signal remains at logic 1 during this duration:

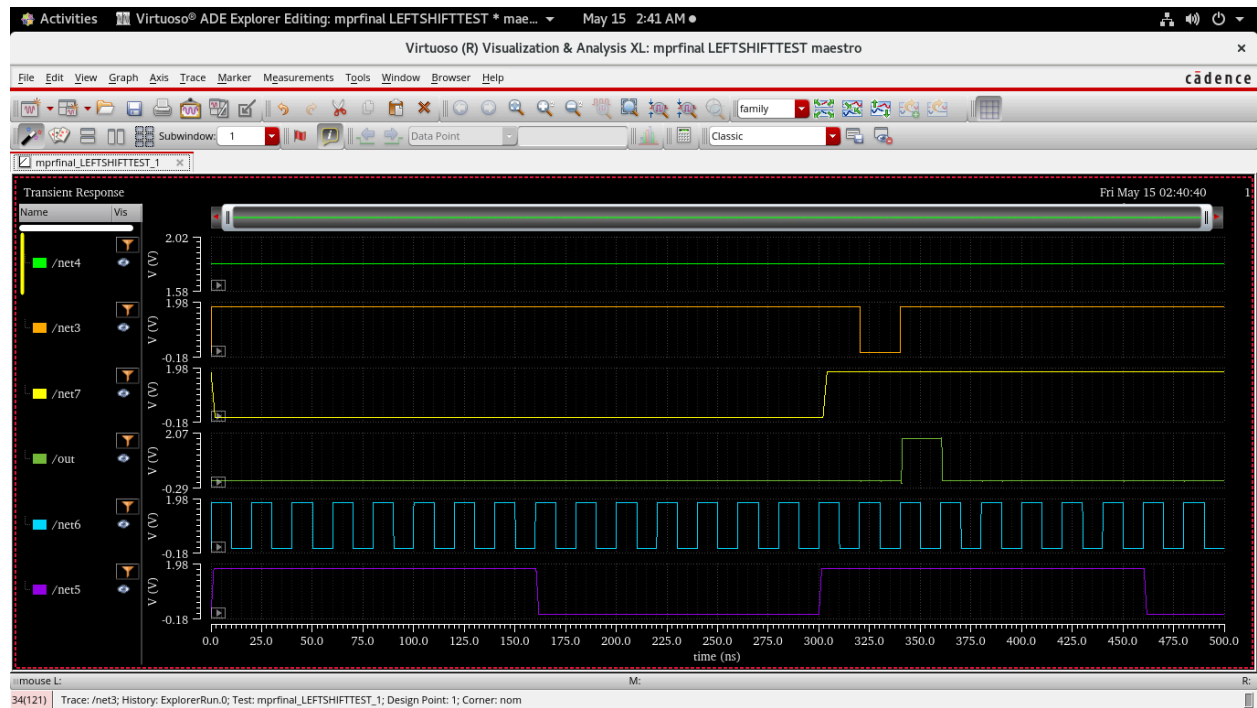
$$159 + 140 \approx 300 \text{ ns} + 20 \text{ ns (For setup time of ENABLE)} = 320 \text{ ns}$$

- After the shifted output becomes available, the **LOADSTORE** signal is changed to logic 0 for one clock cycle (20 ns) so that the PISO register can load the shifted data.
- Once the data is loaded, the PISO register returns to shift mode, and the **LOADSTORE** signal becomes logic 1 again.

8-bit Left Shift Register



OVERALL TIMING DIAGRAM:



7.3. Analysis time

- The total analysis time is calculated as:

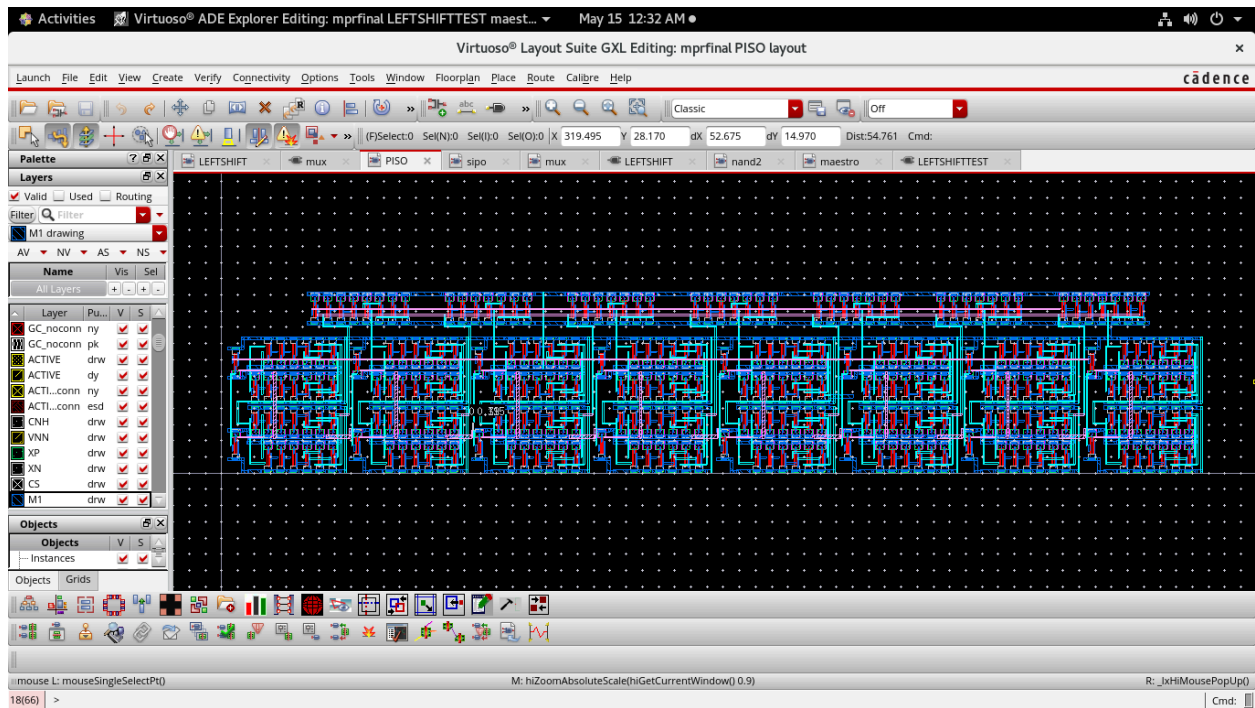
$$320+170=490 \text{ ns}$$

- Therefore, the simulation analysis time is taken as:

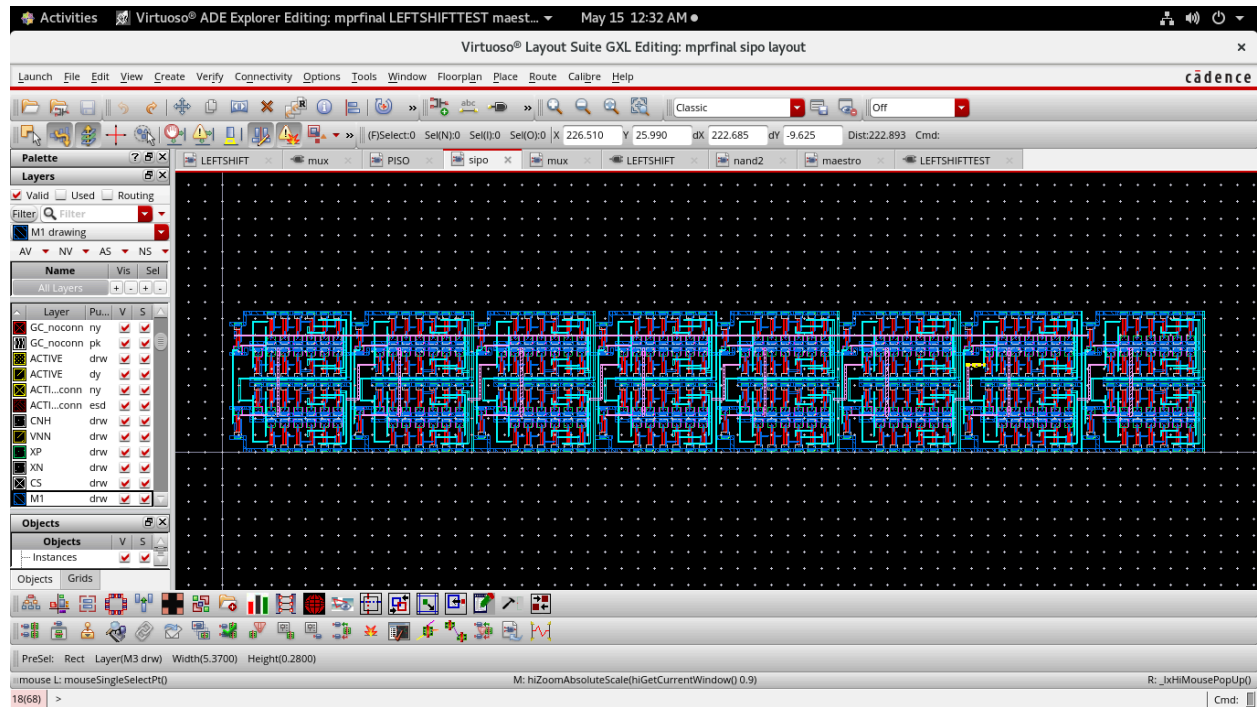
$$500 \text{ ns}$$

10 | LAYOUT

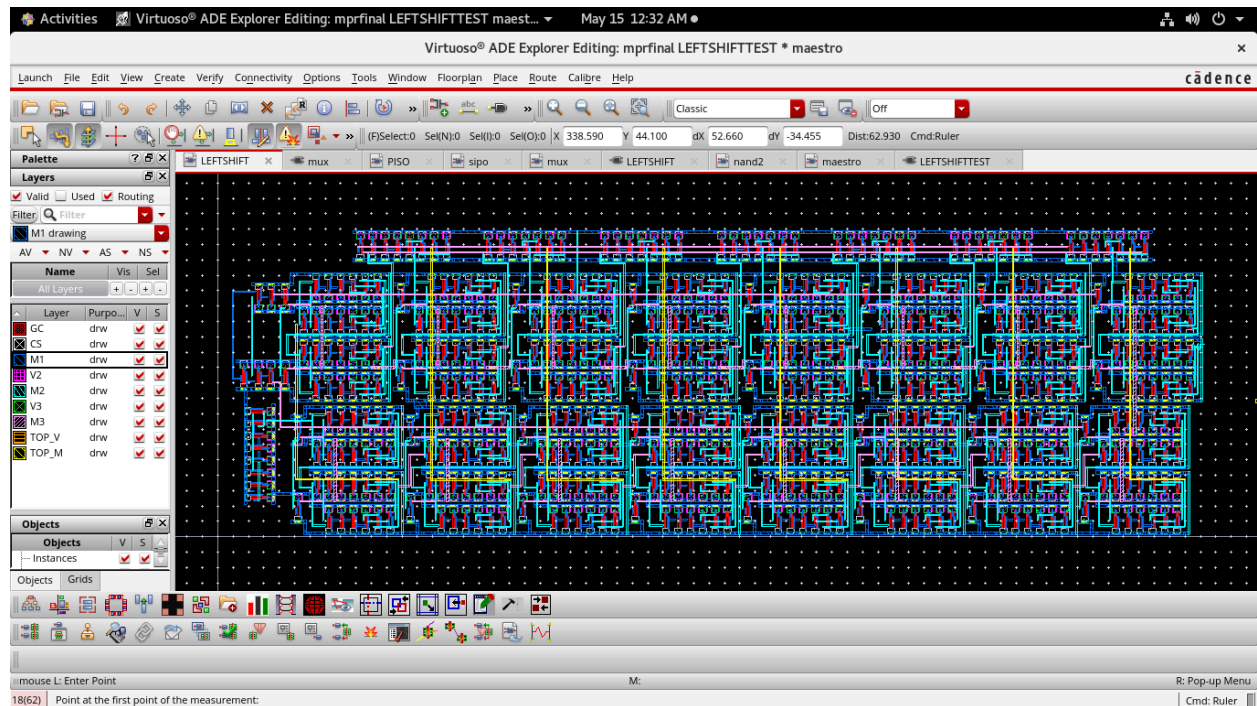
PISO:



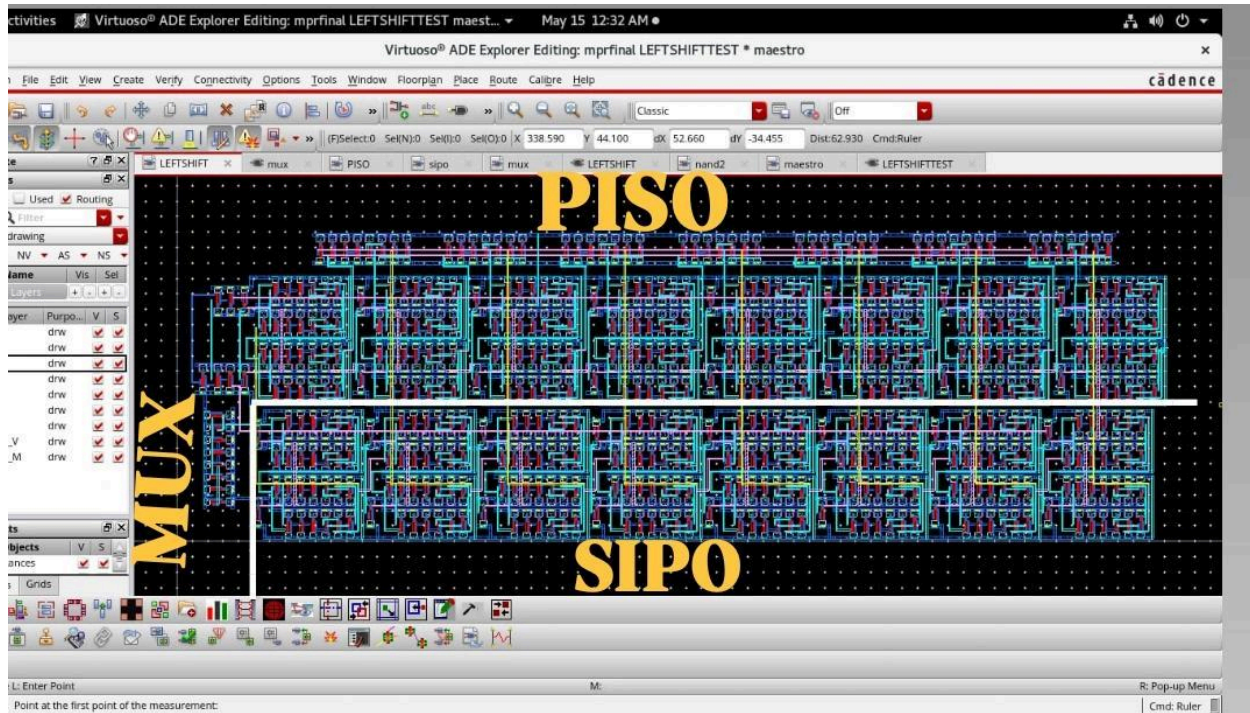
SIPO:



FINAL LAYOUT:



LAYOUT BREAKDOWN:



11 | AREA ANALYSIS

$$\text{Height} = 98.85 \mu\text{m}$$

$$\text{Width} = 316.06 \mu\text{m}$$

$$\text{Area} = \text{Height} \times \text{Width}$$

$$\text{Area} = 98.85 \mu\text{m} \times 316.06 \mu\text{m}$$

$$\text{Area} = 3.124 \times 10^4 \times \mu\text{m}^2$$

12 | POST LAYOUT & PRE LAYOUT COMPARISON

- Delay added due to parasitic capacitance & resistance: 0.5003 ns
- Pre Layout delay = 0.74 ns
- Post Layout delay = 0.74 + 0.5003 = 1.2407 ns

12.1. Post Layout Power Analysis

Parameter	Symbol	Value	Unit
Average Current	I_{avg}	103.2	μA
Peak Current	I_{peak}	3.616	mA
Average Power	P_{avg}	185.76	μW

- Average Power Consumption:

$$\text{Average Power} = V_{DD} \times I_{avg}$$

$$\text{Average Power} = 1.8 \text{ V} \times 103.2 \mu A$$

$$\text{Average Power} = 185.76 \mu W$$

- Energy Consumption:

$$\text{Energy} = T_{sim} \times P_{avg}$$

$$\text{Energy} = 500 \text{ ns} \times 185.76 \mu W$$

$$\text{Energy} = 92.88 \mu J$$